

# Migration Validation & Risk Audit

**Client:** Compliance test  
**Migration:** a to b  
**Audit Date:** 2026-02-03  
**Auditor:** Independent Migration Audit

## Executive Summary

**Final Verdict: NO-GO**

- **Data Volume Checks:** FAIL
- **Aggregate Checks:** FAIL
- **Mapping Checks:** FAIL
- **Data Constraint Checks:** FAIL
- **Relationship Checks:** FAIL

## Data Volume Checks

| Check Name                | Status | Message                                                                                                      |
|---------------------------|--------|--------------------------------------------------------------------------------------------------------------|
| Volume Check: users       | FAIL   | Row count difference exceeds tolerance for table 'users'. Source: 102, Target: 100, Difference: 1.96%.       |
| Volume Check: orders      | FAIL   | Row count difference exceeds tolerance for table 'orders'. Source: 102, Target: 100, Difference: 1.96%.      |
| Volume Check: order_items | FAIL   | Row count difference exceeds tolerance for table 'order_items'. Source: 102, Target: 100, Difference: 1.96%. |

|                        |      |                                                                                                           |
|------------------------|------|-----------------------------------------------------------------------------------------------------------|
| Volume Check: products | FAIL | Row count difference exceeds tolerance for table 'products'. Source: 102, Target: 100, Difference: 1.96%. |
|------------------------|------|-----------------------------------------------------------------------------------------------------------|

Aggregate Checks

| Check Name                  | Status | Message                                                                                                                                          |
|-----------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Sum Check: users - age      | FAIL   | Sum difference exceeds tolerance for column 'age' in table 'users'. Source: 3634, Target: 3559, Difference: 2.06%.                               |
| Average Check: users - age  | WARN   | Average difference within tolerance for column 'age' in table 'users'. Source: 35.627450980392155, Target: 35.59, Difference: 0.11%.             |
| Max Check: users - age      | PASS   | Max matches exactly for column 'age' in table 'users'. Source and Target both have max 41.                                                       |
| Min Check: users - age      | PASS   | Min matches exactly for column 'age' in table 'users'. Source and Target both have min 30.                                                       |
| Variance Check: users - age | WARN   | Variance difference within tolerance for column 'age' in table 'users'. Source: 9.364783537177246, Target: 9.436262626262627, Difference: 0.76%. |
| Sum Check: orders - amount  | FAIL   | Sum difference exceeds tolerance for column 'amount' in table 'orders'. Source: 20280.98, Target: 19875.98, Difference: 2.00%.                   |

|                                  |      |                                                                                                                                                                       |
|----------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Average Check: orders - amount   | WARN | Average difference within tolerance for column 'amount' in table 'orders'.<br>Source:<br>198.83313725490197,<br>Target:<br>198.75979999999998,<br>Difference: 0.04%.  |
| Max Check: orders - amount       | PASS | Max matches exactly for column 'amount' in table 'orders'. Source and Target both have max 330.0.                                                                     |
| Min Check: orders - amount       | PASS | Min matches exactly for column 'amount' in table 'orders'. Source and Target both have min 50.0.                                                                      |
| Variance Check: orders - amount  | FAIL | Variance difference exceeds tolerance for column 'amount' in table 'orders'.<br>Source:<br>6618.199209862164,<br>Target:<br>6684.8302080404055,<br>Difference: 1.01%. |
| Sum Check: orders - quantity     | FAIL | Sum difference exceeds tolerance for column 'quantity' in table 'orders'.<br>Source: 236, Target: 231,<br>Difference: 2.12%.                                          |
| Average Check: orders - quantity | WARN | Average difference within tolerance for column 'quantity' in table 'orders'.<br>Source:<br>2.3137254901960786,<br>Target: 2.31, Difference:<br>0.16%.                 |
| Max Check: orders - quantity     | PASS | Max matches exactly for column 'quantity' in table 'orders'. Source and Target both have max 5.                                                                       |
| Min Check: orders - quantity     | PASS | Min matches exactly for column 'quantity' in table 'orders'. Source and Target both have min 1.                                                                       |

|                                       |      |                                                                                                                                                           |
|---------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variance Check: orders - quantity     | FAIL | Variance difference exceeds tolerance for column 'quantity' in table 'orders'. Source: 1.7025820229081736, Target: 1.7312121212121216, Difference: 1.68%. |
| Sum Check: order_items - price        | FAIL | Sum difference exceeds tolerance for column 'price' in table 'order_items'. Source: 11173.05, Target: 10933.05, Difference: 2.15%.                        |
| Average Check: order_items - price    | WARN | Average difference within tolerance for column 'price' in table 'order_items'. Source: 109.53970588235293, Target: 109.33049999999999, Difference: 0.19%. |
| Max Check: order_items - price        | PASS | Max matches exactly for column 'price' in table 'order_items'. Source and Target both have max 210.5.                                                     |
| Min Check: order_items - price        | PASS | Min matches exactly for column 'price' in table 'order_items'. Source and Target both have min 50.0.                                                      |
| Variance Check: order_items - price   | WARN | Variance difference within tolerance for column 'price' in table 'order_items'. Source: 816.7939771403611, Target: 818.413934090909, Difference: 0.20%.   |
| Sum Check: order_items - quantity     | FAIL | Sum difference exceeds tolerance for column 'quantity' in table 'order_items'. Source: 170, Target: 174, Difference: 2.35%.                               |
| Average Check: order_items - quantity | FAIL | Average difference exceeds tolerance for column 'quantity' in table 'order_items'. Source: 1.666666666666667, Target: 1.74, Difference: 4.40%.            |

|                                        |      |                                                                                                                                                                  |
|----------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Max Check: order_items - quantity      | PASS | Max matches exactly for column 'quantity' in table 'order_items'. Source and Target both have max 4.                                                             |
| Min Check: order_items - quantity      | WARN | Source min for column 'quantity' in table 'order_items' is zero.                                                                                                 |
| Variance Check: order_items - quantity | FAIL | Variance difference exceeds tolerance for column 'quantity' in table 'order_items'. Source: 0.5808580858085808, Target: 0.47717171717171736, Difference: 17.85%. |
| Sum Check: products - price            | WARN | Sum difference within tolerance for column 'price' in table 'products'. Source: 10867.98, Target: 10838.0, Difference: 0.28%.                                    |
| Average Check: products - price        | FAIL | Average difference exceeds tolerance for column 'price' in table 'products'. Source: 106.54882352941176, Target: 108.38, Difference: 1.72%.                      |
| Max Check: products - price            | PASS | Max matches exactly for column 'price' in table 'products'. Source and Target both have max 999.99.                                                              |
| Min Check: products - price            | PASS | Min matches exactly for column 'price' in table 'products'. Source and Target both have min 3.99.                                                                |
| Variance Check: products - price       | FAIL | Variance difference exceeds tolerance for column 'price' in table 'products'. Source: 26291.31828771113, Target: 26649.21, Difference: 1.36%.                    |
| Sum Check: products - stock_quantity   | FAIL | Sum difference exceeds tolerance for column 'stock_quantity' in table 'products'. Source: 29735, Target: 27935, Difference: 6.05%.                               |

|                                           |      |                                                                                                                                                                 |
|-------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Average Check: products - stock_quantity  | FAIL | Average difference exceeds tolerance for column 'stock_quantity' in table 'products'. Source: 291.51960784313724, Target: 279.35, Difference: 4.17%.            |
| Max Check: products - stock_quantity      | PASS | Max matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have max 2000.                                                      |
| Min Check: products - stock_quantity      | PASS | Min matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have min 25.                                                        |
| Variance Check: products - stock_quantity | FAIL | Variance difference exceeds tolerance for column 'stock_quantity' in table 'products'. Source: 86184.05406717143, Target: 78477.60353535356, Difference: 8.94%. |

Mapping Checks

| Check Name                 | Status | Message                                                                                                                                |
|----------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------|
| Mapping Check: users       | FAIL   | Mapping issues found in table 'users': Column 'status' in table 'users' has invalid values: ['FR' 'JP' 'CN' 'US' 'UK' 'CA' 'AU' 'DE']. |
| Mapping Check: users       | FAIL   | Mapping issues found in table 'users': Column 'country_code' in table 'users' has invalid values: [nan].                               |
| Mapping Check: orders      | PASS   | All mappings are valid for table 'orders'.                                                                                             |
| Mapping Check: orders      | PASS   | All mappings are valid for table 'orders'.                                                                                             |
| Mapping Check: order_items | PASS   | All mappings are valid for table 'order_items'.                                                                                        |

|                         |      |                                              |
|-------------------------|------|----------------------------------------------|
| Mapping Check: products | PASS | All mappings are valid for table 'products'. |
| Mapping Check: products | PASS | All mappings are valid for table 'products'. |

Data Constraint Checks

| Check Name                                     | Status | Message                                                                                                           |
|------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------|
| Data Constraints Check: users.first_name       | PASS   | All data constraints are satisfied for table 'users'.                                                             |
| Data Constraints Check: users.last_name        | PASS   | All data constraints are satisfied for table 'users'.                                                             |
| Data Constraints Check: users.email            | PASS   | All data constraints are satisfied for table 'users'.                                                             |
| Data Constraints Check: users.date_of_birth    | PASS   | All data constraints are satisfied for table 'users'.                                                             |
| Data Constraints Check: users.date_onboarded   | FAIL   | Data constraint issues found in table 'users': Column 'date_onboarded' has 1 invalid date values out of 100 rows. |
| Data Constraints Check: orders.order_date      | PASS   | All data constraints are satisfied for table 'orders'.                                                            |
| Data Constraints Check: orders.amount          | PASS   | All data constraints are satisfied for table 'orders'.                                                            |
| Data Constraints Check: orders.user_id         | PASS   | All data constraints are satisfied for table 'orders'.                                                            |
| Data Constraints Check: order_items.order_id   | PASS   | All data constraints are satisfied for table 'order_items'.                                                       |
| Data Constraints Check: order_items.product_id | PASS   | All data constraints are satisfied for table 'order_items'.                                                       |
| Data Constraints Check: order_items.quantity   | PASS   | All data constraints are satisfied for table 'order_items'.                                                       |

|                                                    |      |                                                             |
|----------------------------------------------------|------|-------------------------------------------------------------|
| Data Constraints Check:<br>order_items.price       | PASS | All data constraints are satisfied for table 'order_items'. |
| Data Constraints Check:<br>products.product_name   | PASS | All data constraints are satisfied for table 'products'.    |
| Data Constraints Check:<br>products.price          | PASS | All data constraints are satisfied for table 'products'.    |
| Data Constraints Check:<br>products.stock_quantity | PASS | All data constraints are satisfied for table 'products'.    |

Relationship Checks

| Check Name                             | Status | Message                                                                                                         |
|----------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------|
| Foreign Key Check: orders              | PASS   | All foreign key values in 'user_id' of child table 'orders' have matching primary keys in parent table.         |
| Foreign Key check (Missing FK): orders | FAIL   | Foreign key column 'order_id' not found in child table 'orders'.                                                |
| Foreign Key Check: order_items         | PASS   | All foreign key values in 'order_id' of child table 'order_items' have matching primary keys in parent table.   |
| Foreign Key Check: order_items         | PASS   | All foreign key values in 'product_id' of child table 'order_items' have matching primary keys in parent table. |

Final Deployability Verdict

NO-GO